

第二周作业

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1. 习题一:

6.

由题意得

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.625$$

$$P(\overline{AB}) = P(B) - P(AB) = 0.375$$

$$P(\overline{AB}) = 1 - P(AB) = 0.875$$

$$P((A \cup B)(\overline{AB})) = P(A \cup B) - P(AB) = 0.5$$

11.

根据开车时间把 9:00 到 9:20 分为四段, 则令 $A =$ “甲乙同乘一班车”, $B_i =$ “甲在第 i 段时间内到”, $C_i =$ “乙在第 i 段时间内到”, 则 $P(B_i) = P(C_i) = 0.25$ ($i = 1, 2, 3, 4$) 且甲乙到的时间彼此独立

$$(1) P(A) = P(B_1 C_1 \cup B_2 C_2 \cup B_3 C_3 \cup B_4 C_4) = P(B_1 C_1) + P(B_2 C_2) + P(B_3 C_3) + P(B_4 C_4) = 0.5$$

$$(2) P(A) = P(B_1(C_1 \cup C_2) \cup B_2 \overline{C_4} \cup B_3 \overline{C_1} \cup B_4(C_3 \cup C_4)) = P(B_1)(P(C_1) + P(C_2)) + P(B_2)(1 - P(C_4)) + P(B_3)(1 - P(C_1)) + P(B_4)(P(C_3) + P(C_4)) = 0.625$$

12.

如图, 当针落在阴影区域时原点到该点的连线与 x 轴夹角小于 $\frac{\pi}{4}$

由题意得其为几何概型, 令 $A =$ “原点到该点的连线与 x 轴夹角小于 $\frac{\pi}{4}$ ”

$$P(A) = \frac{S(A)}{S(\Omega)} = \frac{\frac{\pi a^2}{4} + \frac{a^2}{2}}{\frac{\pi a^2}{2}} = \frac{2+\pi}{2\pi}$$

2 14.

$$(1) P(A) = P(AB) + P(\overline{A}B) = 0.2$$

$$P(B) = P(AB) + P(\overline{A}B) = 0.15$$

$$(2) P(B|A) = \frac{P(AB)}{P(A)} = \frac{0.05}{0.2} = 0.25$$

$$(3) P(B|\overline{A}) = \frac{P(\overline{A}B)}{P(\overline{A})} = \frac{0.1}{0.8} = 0.125$$

$$(4) P(A|\overline{B}) = \frac{P(A\overline{B})}{P(\overline{B})} = \frac{0.15}{0.85} = \frac{3}{17}$$

$$(5) P(A|B) = \frac{P(AB)}{P(B)} = \frac{0.05}{0.15} = \frac{1}{3}$$

17.

令 A = “轴承经三次实验未损坏”， B_i = “第 i 次试验中被损坏”，则由题意得

$$P(B_1) = 0.01, P(B_2|\overline{B_1}) = 0.08, P(B_3|\overline{B_1} \cup \overline{B_2}) = 0.15$$

$$P(A) = P(\overline{B_1} \cdot \overline{B_2} \cdot \overline{B_3}) = P(\overline{B_1})P(\overline{B_2}|\overline{B_1})P(\overline{B_3}|\overline{B_1} \cdot \overline{B_2}) = (1 - P(B_1))(1 - P(B_2|\overline{B_1}))(1 - P(B_3|\overline{B_1} \cup \overline{B_2})) = 0.77418$$

20.

令 A = “套中一次”， B_1, B_2, B_3 分别为事件 “选中甲”、“选中乙”、“选中丙”，由题意得

$$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$$

$$P(A|B_1) = C_3^1 \times 0.1 \times (1 - 0.1)^2 = 0.243$$

$$P(A|B_2) = C_3^1 \times 0.2 \times (1 - 0.2)^2 = 0.384$$

$$P(A|B_3) = C_3^1 \times 0.3 \times (1 - 0.3)^2 = 0.441$$

$$\text{故 } P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) = 0.356$$

$$(2) P(B_3|A) = \frac{P(A|B_3)P(B_3)}{P(A)} = 0.4129$$

21.

(1) 令 $A =$ “目标被摧毁”, $B_i =$ “目标被击中 i 次”

$$P(B_1) = C_3^1 \times 0.4 \times (1 - 0.4)^2 = 0.432$$

$$P(B_2) = C_3^2 \times 0.4^2 \times (1 - 0.4) = 0.288$$

$$P(B_3) = C_3^3 \times 0.4^3 = 0.064$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) = 0.2816$$

$$(2) P(B_3|A) = \frac{P(A|B_3)P(B_3)}{P(A)} = 0.1818$$

2. 1.2-5 补充题一:

(1) 由题意得

$$\begin{aligned} P(A \cup B)P(AB) &= (P(A - AB) + P(AB) + P(B - AB))P(AB) \leq P(A - AB)P(B - AB) + (P(A - AB) + P(AB) + P(B - AB))P(AB) \\ &= (P(A - AB) + P(AB))(P(B - AB) + P(AB)) = P(A)P(B) \\ \therefore P(A)P(B) &\geq P(A \cup B)P(AB) \end{aligned}$$

1.3-6 补充题一:

(1) 令 $A =$ “此人患肝癌”, $B_1 =$ “甲胎蛋白低于 200”, $B_2 =$ “甲胎蛋白在 200 与 300 之间”, $B_3 =$ “甲胎蛋白高于 300”。

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) = 0.0875$$

$$(2) P(B_2|A) = \frac{P(A|B_2)P(B_2)}{P(A)} = 0.2857$$